

StockTrader: Comparative Analysis Report

1. Strategy Selection and Rationale

This system pairs the Momentum Trader against the Value Contrarian. These two strategies were chosen because they represent fundamentally opposing investment philosophies that interpret the same market data in opposite ways, maximizing the chance of producing meaningful disagreements for analysis.

Momentum Trader: "The trend is your friend." This strategy follows rising prices, bullish moving average crossovers (20-day MA above 50-day MA), increasing volume, and consistent positive daily returns. When a stock is trending upward, it buys; when trending downward, it sells. It never tries to catch a falling knife.

Value Contrarian: "Markets overreact. Buy fear, sell greed." This strategy interprets sharp price drops as buying opportunities and sharp surges as overvaluation risks. It relies on distance from the 52-week high/low, RSI levels, and recent drawdowns to identify mispriced stocks.

The expected pattern of disagreement was clear before running the system: when a stock is declining, the Momentum Trader should want to sell (following the downward trend) while the Value Contrarian should want to buy (interpreting the drop as an overreaction). When a stock is surging, the Momentum Trader should buy while the Contrarian should sell. Agreement was expected only in extreme or ambiguous cases.

2. System Architecture

The system is built in plain Python using `concurrent.futures.ThreadPoolExecutor` for parallel strategy execution. No external agent framework is used. The architecture follows the required workflow with four components: Market Data, two Strategy Agents, and an Evaluator, plus the optional Debate Mode extension.



3. Stock Selection and Rationale

Ticker	Category	Rationale
AAPL	Steady large-cap	Apple is the quintessential blue-chip stock. A modest 3.27% decline over 30 days with low volatility (0.013) makes it an ideal baseline for comparison.
NVDA	Volatile high-momentum	NVIDIA is at the center of the AI boom with high volatility (0.0243). At 16.39% below its 52-week high with a -6.54% monthly decline, it creates a textbook disagreement scenario.
INTC	Recently declined	Intel has faced competitive pressure. Despite a recent 14.21% bounce, its 52-week range (\$17.67-\$54.60) and 24.17% max drawdown reflect extreme instability.
KO	Trading sideways	Coca-Cola is a defensive consumer staple with the lowest volatility (0.009). Only 5.8% below its 52-week high with a neutral RSI of 50.09, it represents the ambiguous middle ground.

4. Results by Stock

Ticker	Momentum Trader	Conf.	Value Contrarian	Conf.	Result
AAPL	SELL	8	HOLD	6	DISAGREE
NVDA	SELL	8	BUY	8	DISAGREE
INTC	SELL	7	SELL	7	AGREE
KO	SELL	8	HOLD	6	DISAGREE

AAPL — Apple Inc.

AAPL traded at \$255.92, down 3.27% over 30 days with its 20-day MA (\$253.41) below the 50-day MA (\$260.30). The Momentum Trader issued a confident SELL (8/10), citing the bearish moving average crossover and negative average daily return of -0.1065%. The Value Contrarian recommended HOLD (6/10), noting that the stock is only 11.25% below its 52-week high and the RSI of 57.46 sits in neutral territory — not enough to signal either a buy or sell opportunity. The evaluator identified the root cause: the Momentum Trader focuses on short-term price direction and sees a clear downtrend, while the Contrarian needs a much larger drop to trigger a buy signal.

NVDA — NVIDIA Corp.

NVDA at \$177.39 showed the sharpest disagreement: SELL (8/10) from the Momentum Trader vs. BUY (8/10) from the Value Contrarian — a direct clash with equal confidence. The following excerpts from the JSON output files illustrate the opposing reasoning:

- Momentum Trader: "The 20-day moving average at \$177.62 is below the 50-day moving average of \$182.64, indicating a bearish signal. Additionally, the 30-day price change of -6.54% reflects negative momentum, while the volume trend is decreasing, which suggests weakening demand."
- Value Contrarian: "NVDA is currently priced at \$177.39, which is 16.39% below its 52-week high of \$212.17, indicating that it may be oversold. The stock has experienced a max drawdown of 15.54%, signaling fear in the market, which aligns with my contrarian buying strategy."

This is the textbook case of the same data producing opposite conclusions. The 6.54% decline and bearish MA crossover are sell signals for the trend-follower but buy signals for the contrarian. Neither agent changed its position during the Debate Mode, both maintaining 8/10 confidence.

INTC — Intel Corp.

INTC was the only stock where both strategies agreed — both recommended SELL (7/10). This is noteworthy because INTC had actually risen 14.21% over 30 days. The Momentum Trader sold because the longer-term moving average structure was still bearish (20-day MA at \$45.29 below 50-day MA at \$46.16) despite the recent bounce. The Value Contrarian sold because a 14.21% surge in 30 days triggers its "sell greed" instinct, with the stock only 7.73% below its 52-week high and RSI nearing overbought at 62.11. Different reasons, same conclusion — a genuine convergence.

KO — Coca-Cola Co.

KO at \$76.72 showed a mild disagreement: SELL (8/10) from the Momentum Trader vs. HOLD (6/10) from the Value Contrarian. The stock's 20-day MA (\$76.24) sat just barely below the 50-day MA (\$76.71) — a difference of only \$0.47. The Momentum Trader interpreted this as a bearish crossover signal, while the Value Contrarian saw a stock only 5.8% off its 52-week high with a neutral RSI of 50.09 — too close to fair value to trigger a strong signal in either direction.

5. Patterns of Agreement and Disagreement

Across four stocks, the strategies agreed on only one (INTC) and disagreed on three (AAPL, NVDA, KO). This 1:3 ratio confirms that these two philosophies genuinely interpret market data differently, rather than being cosmetic variations of the same approach.

When they disagree: The pattern is consistent. In all three disagreements, the Momentum Trader recommended SELL while the Value Contrarian recommended HOLD or BUY. The current market environment (early April 2026) showed broad softness across equities, with most stocks experiencing modest declines. For the Momentum Trader, any downtrend — however mild — triggers a sell signal. For the Value Contrarian, a mild decline is simply noise that does not warrant action unless the stock is deeply oversold.

The sharpness of disagreement correlated with the magnitude of the decline and the distance from the 52-week high. NVDA, with a 6.54% drop and 16.39% distance from its high, produced the sharpest clash (SELL vs. BUY). AAPL and KO, with smaller ~3% declines and modest distances from their highs, produced milder disagreements (SELL vs. HOLD).

When they agree: INTC was the sole agreement, and it was the most counterintuitive case. Despite a strong 14.21% monthly gain, both agents recommended SELL. The Momentum Trader saw bearish structure beneath the surface rally (MAs still crossed bearishly), while the Contrarian saw a sharp surge that triggered its "sell greed" instinct. This demonstrates that agreement between opposing philosophies often signals a stock in an unusual or contradictory state — the kind that should prompt extra caution.

Debate Mode insights: In all three disagreement cases, both agents held firm during the debate round. No positions changed, and confidence levels remained stable (the Value Contrarian slightly increased confidence from 6 to 7 on AAPL but did not change its HOLD decision). This suggests the behavioral philosophies are deeply embedded in the prompt design and resistant to persuasion, which is realistic, as real investors with strong philosophical commitments rarely change their minds from a single counter-argument.

6. Failure or Surprise Case

The most surprising result was the Momentum Trader recommending SELL on every single stock, including INTC, which had risen 14.21% in 30 days with increasing volume. A true momentum trader should have been excited by INTC's strong upward move. However, the agent fixated on the moving average crossover (20-day MA below 50-day MA) and dismissed the strong recent price action.

This reveals a limitation in the prompt design: the system prompt weighted the MA crossover signal too heavily relative to other momentum indicators. A more sophisticated momentum trader would recognize that INTC's 14.21% surge and increasing volume represent genuine momentum even if the longer-term MA structure has not yet caught up. Moving averages are lagging indicators by definition — they take time to reflect a trend reversal. The LLM over-indexed on one signal at the expense of the holistic picture, producing a decision that a human momentum trader would likely disagree with.

7. Reflection

If I had \$10,000 and had to follow one strategy for the next month, I would choose the Value Contrarian. In a market environment with broad softness but no crash, the Momentum Trader's "sell everything" tendency leads to premature exits. The Contrarian's patience — holding through modest declines and buying on significant drops — is better suited to choppy, range-bound markets where sharp drops tend to recover.

A hybrid strategy that combines the best of both would use the Momentum Trader's moving average analysis to identify trend direction, but apply the Value Contrarian's threshold discipline to filter out noise. For example: only sell when the MA crossover is bearish, AND the stock has dropped more than 10% from its recent peak; only buy when the stock is more than 15% below its 52-week high, AND the RSI is below 35. This would reduce false signals from both sides while capturing the genuine insights each philosophy offers.

8. Extension: Debate Mode

I implemented the Debate Mode extension, which triggers a second round of analysis whenever the Momentum Trader and Value Contrarian disagree. In this round, each agent receives the opposing agent's decision, confidence, and justification, then has the opportunity to revise its own position.

Across the three disagreement cases (AAPL, KO, NVDA), no agent changed its decision after seeing the other's reasoning. However, the debate round did reveal meaningful shifts in confidence and argumentation. For NVDA, both agents held firm at confidence 8, each doubling down on their framework — the Momentum Trader citing continued downward trend signals, the Value Contrarian arguing the drawdown represented a buying opportunity. For AAPL and KO, the Value Contrarian slightly lowered its confidence after acknowledging the Momentum Trader's trend concerns, but still maintained its original position.

This suggests that the agents' behavioral philosophies are deeply embedded in their prompts rather than superficially held. When confronted with opposing evidence, they incorporated it into their reasoning but interpreted it through their own strategic lens — much like real investors who acknowledge counterarguments without abandoning their framework. The debate mode confirmed that the disagreements in our system are genuine philosophical divergences, not artifacts of weak prompting.

AI Use Appendix

1. AI Tools Used During Development

Claude Code (Anthropic Claude Opus 4.6)

2. Representative Prompts Used During Development

"Read the assignment instructions very carefully. Understand all the requirements and detailed instructions. For the Stock selection, choose AAPL, NVDA, INTC, and KO. For the strategy menu, I want you to choose two strategies that fit the most (except News Sentiment Follower). Start to work on this assignment, step by step."

"Check the instruction PDF, I see there is a 'Optional Extension Bonus' section. I want to add a debate mode. Work on this right now."

3. Representative Output

Claude Code — Generated market_data.py snippet

```
def fetch_all_market_data(tickers, max_retries=5):
    raw = yf.download(
        tickers,
        start=start_date.strftime("%Y-%m-%d"),
        end=end_date.strftime("%Y-%m-%d"),
        progress=False,
        group_by="ticker",
    )
    # ... compute indicators per ticker ...
```

4. What Was Accepted, Revised, Rejected, or Verified

Accepted

The overall project structure (src/, prompts/, outputs/, report/ directories) was accepted. The strategy prompt designs for both the Momentum Trader and Value Contrarian were accepted with minor review.

Revised

The market data fetching approach went through multiple revisions. The initial implementation used `yf.Ticker().history()` for individual stocks, which triggered Yahoo Finance rate limiting. This was revised to use `yf.download()` with batch fetching for all tickers in a single API call.

Verified Independently

All JSON output files were reviewed manually to verify that: (1) agent justifications reference specific numerical data from the market data summary rather than generic statements, (2) the evaluator analysis explains root causes of agreement/disagreement rather than merely restating

decisions, (3) the debate mode outputs reference specific points from the opponent's argument, and (4) the market data values (prices, moving averages, RSI, etc.) are consistent between the market_data_summary and the agent justifications. The stock selections were independently verified to represent the four required categories (steady large-cap, volatile high-momentum, recently declined, and sideways-trading).